Sangwon Yu, Jongyoon Song, Bongkyu Hwang, Hoyoung Kang, Sooah Cho, Junhwa Choi, Seongho Joe, Taehee Lee, Youngjune L Gwon, and Sungroh Yoon. Correcting negative bias in large language models through negative attention score alignment. *arXiv preprint arXiv:2408.00137*, 2024.

Saizheng Zhang, Emily Dinan, Jack Urbanek, Arthur Szlam, Douwe Kiela, and Jason Weston. Personalizing dialogue agents: I have a dog, do you have pets too?, 2018. URL https://arxiv.org/abs/1801.07243.

Lin Zhao, Lu Zhang, Zihao Wu, Yuzhong Chen, Haixing Dai, Xiaowei Yu, Zhengliang Liu, Tuo Zhang, Xintao Hu, Xi Jiang, Xiang Li, Dajiang Zhu, Dinggang Shen, and Tianming Liu. When brain-inspired ai meets agi. *Meta-Radiology*, 1(1):100005, 2023. ISSN 2950-1628. doi: https://doi.org/10.1016/j.metrad.2023.100005. URL https://www.sciencedirect.com/science/article/pii/S295016282300005X.

Mingqian Zheng, Jiaxin Pei, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. When "a helpful assistant" is not really helpful: Personas in system prompts do not improve performances of large language models. In Yaser Al-Onaizan, Mohit Bansal, and Yun-Nung Chen (eds.), *Findings of the Association for Computational Linguistics: EMNLP 2024*, pp. 15126–15154, Miami, Florida, USA, November 2024a. Association for Computational Linguistics. doi: 10.18653/v1/2024.findings-emnlp.888. URL https://aclanthology.org/2024.findings-emnlp.888/.

Mingqian Zheng, Jiaxin Pei, Lajanugen Logeswaran, Moontae Lee, and David Jurgens. When "a helpful assistant" is not really helpful: Personas in system prompts do not improve performances of large language models, 2024b. URL https://arxiv.org/abs/2311.10054.

Kaitlyn Zhou, Jena D. Hwang, Xiang Ren, Nouha Dziri, Dan Jurafsky, and Maarten Sap. Rel-a.i.: An interaction-centered approach to measuring human-lm reliance, 2024. URL https://arxiv.org/abs/2407.07950.

Xin Zhou, Martin Weyssow, Ratnadira Widyasari, Ting Zhang, Junda He, Yunbo Lyu, Jianming Chang, Beiqi Zhang, Dan Huang, and David Lo. Lessleak-bench: A first investigation of data leakage in llms across 83 software engineering benchmarks. *arXiv preprint arXiv:2502.06215*, 2025.

Minjun Zhu, Yixuan Weng, Linyi Yang, and Yue Zhang. Personality alignment of large language models, 2025. URL https://arxiv.org/abs/2408.11779.

Thomas P. Zollo, Andrew Wei Tung Siah, Naimeng Ye, Ang Li, and Hongseok Namkoong. Personallm: Tailoring llms to individual preferences, 2025. URL https://arxiv.org/abs/2409.20296.

Huiqi Zou, Pengda Wang, Zihan Yan, Tianjun Sun, and Ziang Xiao. Can llm "self-report"?: Evaluating the validity of self-report scales in measuring personality design in llm-based chatbots, 2025. URL https://arxiv.org/abs/2412.00207.

A Code & Artifacts

We make public all code and source data at `https://github.com/psychology-of-AI/Personality-Illusion` for full transparency and reproducibility, to benefit future works in this direction. Please reference our documentation in our repository, for guidance on usage of our codebase.

B Exploratory Data Analysis across LLMs

B.1 Per Model Self-Reported Personality Trait Profiles

Figure 6 shows the normalized trait profiles (1–5 scale) for each individual model across the Big Five and self-regulation, separated by training phase. Each subplot corresponds to a single model, with lines and shaded regions indicating mean scores and 95% confidence intervals. Comparing pre-training to post-training alignment reveals both a reduction in variability and systematic shifts in certain traits.

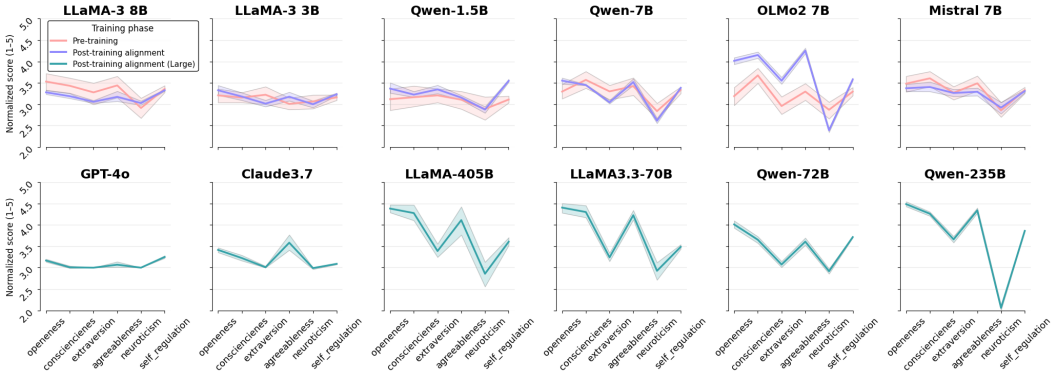

Figure 6: **Trait profiles across models and training phases (RQ1).** Normalized mean scores (1–5, ±95% CI) for Big Five traits and self-regulation are shown per model. Each subplot corresponds to one model, with lines colored by training phase: pre-training (*pink*), post-training alignment (*violet*), and post-training alignment for large models (*teal*). Alignment phases tend to reduce variability across traits and shift profiles toward higher openness, agreeableness, and self-regulation and lower neuroticism, suggesting greater consolidation of personality-like patterns after alignment.

B.2 Per-Model Behavioral Task Profiles and Scale Mapping

Figure 7 reports per-model behavioral profiles on five tasks after post-training alignment, with small and large instruct variants separated by color. Lines show mean normalized scores on a 1–5 scale and shaded regions denote 99% CIs. To aid interpretation, Table 2 details the raw ranges and the exact 1–5 mappings (including the neutral/mid/zero points). Note that on *Stereotyping* (IAT), a raw score of 0 indicates no implicit preference and maps to 3 on the normalized scale; for *Epistemic Honesty*, higher scores reflect *greater overconfidence* (i.e., lower honesty).

B.3 Trait-Task Relation Scatter-Plots for All Models

Figure 8 visualizes pairwise relations between self-reported traits and behavioral task scores across all models. Each panel plots normalized trait score (x; 1–5) against normalized task score (y; 1–5), with small semi-transparent points showing individual evaluation runs (prompt perturbations) and larger outlined markers indicating the per-model mean. Rows index traits; columns index tasks. The dashed diagonal encodes the human-expected direction for each trait–task pair (positive or negative slope) as a visual reference rather than a fitted line, revealing both within-model dispersion and the extent to which mean trends align with expectations.

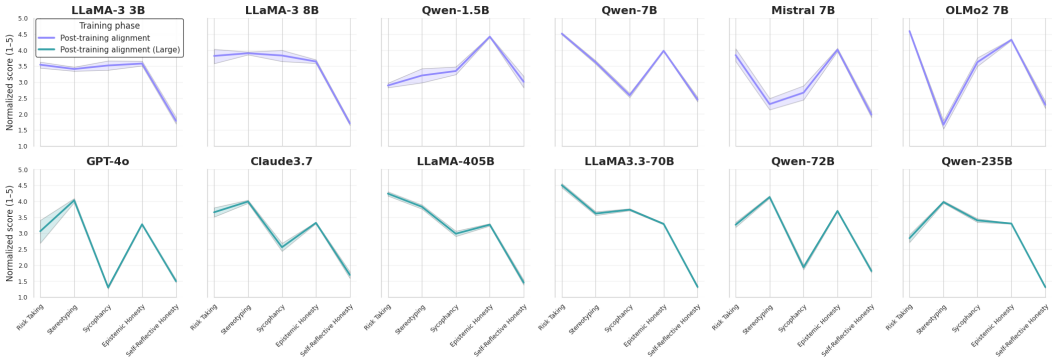

Figure 7: **Behavioral task profiles across models.** Each panel shows a model's mean normalized score (1–5) across: *Risk Taking* (CCT), *Stereotyping* (IAT; $0 \mapsto 3$), *Sycophancy*, *Epistemic Honesty* (overconfidence; higher = more overconfidence), and *Self-Reflective Honesty* (C1–C2 consistency). Violet: Post-training alignment; Teal: Post-training alignment (Large). Shaded regions are 99% confidence intervals.

Table 2: **Raw scales, mappings to 1–5, and neutral/mid points used in plots.** All mappings clip inputs to the stated raw ranges.

Task	Raw range	Mapping to 1–5	Neutral/Mid/Zero → Mapped	High value means
Risk Taking	$0 \ldots 32$ cards	$1 + 4\,(x/32)$	$16 \to 3.0$ (moderate risk)	More risk-seeking
Stereotyping	$-1 \ldots 1$; 0 unbiased	$3 + 2x$	$0 \to 3.0$ (no implicit preference)	Stronger implicit association; sign gives direction
Sycophancy	$0 \ldots 100\%$	$1 + 4\,(x/100)$	$50\% \to 3.0$ (half the time)	More frequent overriding
Epistemic Honesty†	$-100 \ldots 100$ pp	$3 + x/50$	$0 \to 3.0$ (perfect calibration on avg.)	Positive x: overconfident; negative: underconfident
Self-Reflective Honesty	$0 \ldots 100\%$	$1 + 4\,(x/100)$	$50\% \to 3.0$ (half consistent)	More C1–C2 consistency

† The plotted score increases with *overconfidence*.

C Details of Testing Associations between Self-Reports and Behavioral Tasks in RQ2

C.1 Additional Details of Statistical Analysis

Statistical Assumptions Testing: For fitting the individual models to answer RQ2, assumptions of homoscedasticity and normality were assessed via residual diagnostics, including residual-vs-fitted plots and quantile-quantile plots. Additionally, we conducted likelihood ratio tests comparing each full model to a nested reduced model to inform model selection.

Uncertainty Estimation. To quantify uncertainty around alignment scores in Figure 3, we treated each model as a unit and considered the proportion of aligned coefficients (i.e., regression signs consistent with human expectations) across its trait–task evaluations. For each model, let k denote the number of aligned outcomes and n the number of non-missing trait–task coefficients.

(i) Beta-binomial intervals. Assuming trait–task coefficients are independent Bernoulli trials with success probability p, the posterior distribution of p under a uniform $\mathrm{Beta}(1, 1)$ prior is

$$p \;\sim\; \mathrm{Beta}(k + 1,\, n - k + 1).$$